

EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 2B

1. A Special Complex Basis

Recall that vectors can be complex valued, and complex-valued signals with four samples can be represented as vectors in $\mathbb{C}^4$. In this class, we define the inner product between complex vectors $\mathbf{x}, \mathbf{y} \in \mathbb{C}^N$ as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}^* = \sum_{n=0}^{N-1} x[n] y[n]^*$$

where $y[n]^*$ denotes complex conjugation. Consider the vectors

$$\boldsymbol{\phi}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{\phi}_1 = \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix}, \quad \boldsymbol{\phi}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \quad \boldsymbol{\phi}_3 = \begin{bmatrix} 1 \\ -i \\ -1 \\ i \end{bmatrix}.$$

It may also be handy to group them as columns into one matrix:

$$\Phi = \begin{bmatrix} | & | & | & | \\ \boldsymbol{\phi}_0 & \boldsymbol{\phi}_1 & \boldsymbol{\phi}_2 & \boldsymbol{\phi}_3 \\ | & | & | & | \end{bmatrix}.$$

- (a) Compute the Gram matrix G , whose (k, ℓ) entry is $G_{k\ell} = \langle \boldsymbol{\phi}_k, \boldsymbol{\phi}_\ell \rangle$. Are the $\boldsymbol{\phi}_k$ orthogonal or orthonormal? Do they form a valid basis for $\mathbb{C}^4$, and give the corresponding normalized basis.

Hint: what is $\langle \boldsymbol{\phi}_a, \boldsymbol{\phi}_b \rangle$ when $a = b$, and when $a \neq b$? How could Φ help you?

Answer: Orthogonality requires $\langle \boldsymbol{\phi}_k, \boldsymbol{\phi}_\ell \rangle = 0$ for every distinct pair; the diagonal entries instead give the squared norms. For example,

$$\langle \boldsymbol{\phi}_0, \boldsymbol{\phi}_1 \rangle = 1 - i - 1 + i = 0, \quad \langle \boldsymbol{\phi}_1, \boldsymbol{\phi}_1 \rangle = 1 + i(-i) + (-1)^2 + (-i)i = 4.$$

Conjugating the second argument in every entry the same way gives

$$G = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} = 4I_4.$$

So the $\boldsymbol{\phi}_k$ are mutually orthogonal, but since $\|\boldsymbol{\phi}_k\| = \sqrt{4} = 2$ they are not orthonormal. Four nonzero orthogonal vectors are linearly independent, and $\dim(\mathbb{C}^4) = 4$, so they form a basis; dividing each by 2 gives the normalized basis $\{\boldsymbol{\phi}_k/2\}$.

- (b) Suppose

$$\mathbf{x} = \sum_{k=0}^3 \alpha_k \boldsymbol{\phi}_k.$$

Take the inner product of both sides with ϕ_ℓ and derive a formula for the coordinate α_ℓ .

Hint: when you expand $\langle \mathbf{x}, \phi_\ell \rangle$, what does orthogonality do to the terms with $k \neq \ell$?

Answer: Using linearity in the first argument and the orthogonality found in part (a),

$$\begin{aligned}\langle \mathbf{x}, \phi_\ell \rangle &= \sum_{k=0}^3 \alpha_k \langle \phi_k, \phi_\ell \rangle \\ &= 4\alpha_\ell.\end{aligned}$$

Therefore

$$\alpha_\ell = \frac{\langle \mathbf{x}, \phi_\ell \rangle}{\|\phi_\ell\|^2} = \frac{1}{4} \langle \mathbf{x}, \phi_\ell \rangle.$$

(c) Express

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

in the $\{\phi_k\}$ basis. In other words, find the coefficients $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ such that

$$\mathbf{u} = \sum_{k=0}^3 \alpha_k \phi_k = \alpha_0 \phi_0 + \alpha_1 \phi_1 + \alpha_2 \phi_2 + \alpha_3 \phi_3.$$

Then compute the projection of $\mathbf{u}$ onto $\text{span}\{\phi_2\}$ and explain its relationship to the ϕ_2 component of $\mathbf{u}$.

Hint: part (b) gives α_ℓ directly — one inner product per coefficient, no system of equations to solve.

Answer: Applying the coordinate formula gives

$$\begin{aligned}\alpha_0 &= \frac{1}{4} \langle \mathbf{u}, \phi_0 \rangle = 1, & \alpha_1 &= \frac{1}{4} \langle \mathbf{u}, \phi_1 \rangle = -\frac{1}{2}, \\ \alpha_2 &= \frac{1}{4} \langle \mathbf{u}, \phi_2 \rangle = 1, & \alpha_3 &= \frac{1}{4} \langle \mathbf{u}, \phi_3 \rangle = -\frac{1}{2}.\end{aligned}$$

Hence

$$\mathbf{u} = \phi_0 - \frac{1}{2}\phi_1 + \phi_2 - \frac{1}{2}\phi_3.$$

The projection is

$$\text{proj}_{\phi_2}(\mathbf{u}) = \frac{\langle \mathbf{u}, \phi_2 \rangle}{\|\phi_2\|^2} \phi_2 = \frac{4}{4} \phi_2 = \phi_2.$$

This is exactly the ϕ_2 component of $\mathbf{u}$! ($\alpha_2 \phi_2$).

2. More Euler's Formula! (Adapted from EE 120 Homework)

Learning Objectives: Practice complex exponentials by deriving basic properties using Euler's Formula.

Euler's Formula is defined as follows:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

(a) Derive the following identities using Euler's Formula:

i. $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$

Answer: $\cos \theta = \cos \theta + \frac{i}{2} \sin \theta - \frac{i}{2} \sin \theta$
 $= (\frac{1}{2} \cos \theta + \frac{i}{2} \sin \theta) + (\frac{1}{2} \cos \theta - \frac{i}{2} \sin \theta)$
 $= \frac{1}{2} [(\cos \theta + i \sin \theta) + (\cos \theta - i \sin \theta)]$
 $= \frac{1}{2} (e^{i\theta} + e^{-i\theta})$
 $= \frac{e^{i\theta} + e^{-i\theta}}{2}$

ii. $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

Answer: $\sin \theta = \sin \theta + \frac{1}{2i} \cos \theta - \frac{1}{2i} \cos \theta$
 $= (\frac{1}{2i} i \sin \theta + \frac{1}{2i} \cos \theta) + (\frac{1}{2i} i \sin \theta - \frac{1}{2i} \cos \theta)$
 $= \frac{1}{2i} [(\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta)]$
 $= \frac{1}{2i} (e^{i\theta} - e^{-i\theta})$
 $= \frac{e^{i\theta} - e^{-i\theta}}{2i}$

(b) De Moivre's Theorem states the following relationship for any $\theta \in \mathbb{R}, n \in \mathbb{Z}$:

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

Use Euler's formula to derive this relationship.

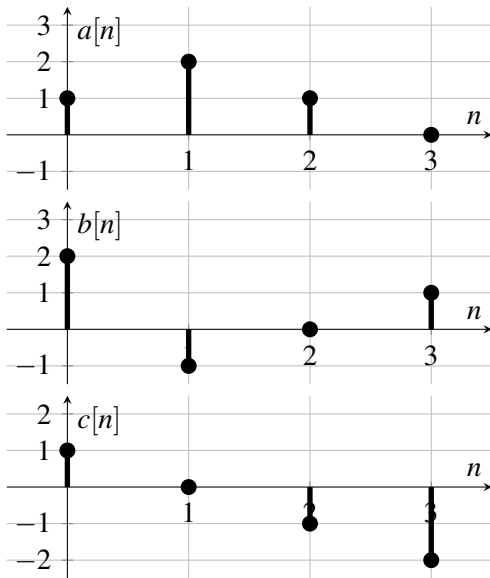
Answer: $(\cos \theta + i \sin \theta)^n = (e^{i\theta})^n = e^{in\theta} = e^{i(n\theta)} = \cos n\theta + i \sin n\theta$

3. (Optional) Signal Operations

As a digital sound designer, you have been contracted by an opera company to help them record and process their vocal performances for release online. To help each singer shine, you will be using your deep knowledge of audio signal processing to distinguish each of the vocal performances and ensure that each singer is on pitch.

- (a) After gathering all of the group's recordings, you realize that you may have mixed up some of the tracks! In order to make sure that you're using the right tracks for each of the singers, make sure that all of recordings are **mutually orthogonal**; this will tell us that the different tracks come from singers with a different pitch.

Hint: Represent each signal as a vector with 4 entries.



Solution: We can begin by expressing each of our signals as length 4 vectors:

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ -2 \end{bmatrix}$$

From here, we can ensure mutual orthogonality by taking the inner products of each pair of vectors:

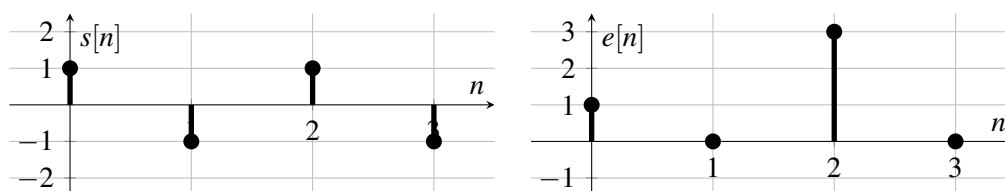
$$\langle \mathbf{a}, \mathbf{b} \rangle = 1(2) + 2(-1) + 1(0) + 0(1) = 2 - 2 + 0 + 0 = 0$$

$$\langle \mathbf{a}, \mathbf{c} \rangle = 1(1) + 2(0) + 1(-1) + 0(-2) = 1 + 0 - 1 + 0 = 0$$

$$\langle \mathbf{b}, \mathbf{c} \rangle = 2(1) + (-1)(0) + 0(-1) + 1(-2) = 2 + 0 + 0 - 2 = 0$$

As we can see, the inner products are all zero, so our voice tracks are in fact mutually orthogonal. There are no duplicates between singers' voices.

- (b) Upon further inspection, you discover that the soprano's microphone was muted the whole time! Instead of having her re-record, you decide to extract her voice by comparing an earlier voice sample with an ensemble recording of the group. To do this, project the ensemble recording (the signal e) onto the basis vector formed by the soprano's voice sample (the signal s).



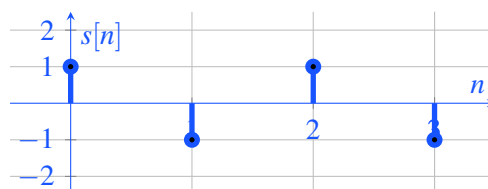
Answer: First, we can begin by expressing our two signals as vectors:

$$\mathbf{s} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

From here, we can project the ensemble recording onto the soprano's voice sample using the projection formula:

$$\begin{aligned} \text{proj}_{\mathbf{s}}(\mathbf{e}) &= \frac{\langle \mathbf{e}, \mathbf{s} \rangle}{\|\mathbf{s}\|^2} \mathbf{s} \\ &= \frac{1(1) + 0(-1) + 3(1) + 0(-1)}{1^2 + (-1)^2 + 1^2 + (-1)^2} \mathbf{s} \\ &= \frac{1 + 0 + 3 + 0}{1 + 1 + 1 + 1} \mathbf{s} \\ &= \frac{4}{4} \mathbf{s} \\ &= \mathbf{s} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \end{aligned}$$

Therefore, our soprano's voice signal from the ensemble recording is:



As it turns out, the soprano's voice sample was actually her recording all along! We should name our files better next time.